

Report: 2024 Market Analysis & ETL Process

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1. Modeling Choices: For the data architecture, I decided to use a Star Schema design. I set the grain of the central Fact_Transactions table at the lowest level possible, meaning one row represents exactly one financial transaction. Around this fact table, I created four dimension tables: Dim_Time, Dim_Geography, Dim_Symbol, and Dim_TransactionType. To make the database faster and keep the relationships clean, I added surrogate keys (integers) to all tables. I also made sure to remove all the unnecessary text columns from the fact table so it stays lightweight and optimized.

2. Data Quality Issues Found: During the ETL process, I ran into an issue with unmapped data and took specific steps to investigate and structure it:

- Investigating orphan records: When I first joined the account statement with the symbols file, I found 212 "orphan" transactions (symbols that didn't match). I suspected this might be due to typos, hidden spaces, or uppercase/lowercase mix-ups. To test this, I applied the upper() and strip() functions defensively to clean the text. However, the orphan count remained exactly at 212. This proved that these symbols were truly missing from the reference dictionary, and it wasn't just a formatting bug. To maintain strict referential integrity in my Star Schema, I safely dropped these 212 unmapped rows.
- Standardizing country names: To make sure the symbols dataset would join perfectly with the official ISO country dataset, I proactively created a mapping dictionary. This allowed me to replace informal names (like "USA" or "UK") with their exact official ISO equivalents (like "United States of America") before doing the merge.

3. Main Analytical Results: After running the SQL queries on the Star Schema, I found some interesting market trends for 2024:

- Sectors and assets: The Technology sector is by far the most dominant one. It was the top sector for SELL transactions in the US with 158 transactions. Also, the most traded companies overall were ARM and AMD (with 100 and 97 transactions respectively).
- Time trends: The market activity actually dropped a lot over the year. Quarter 1 was the busiest period with 999 transactions, but by Q3 and Q4, the activity went down to only about 260 transactions per quarter.
- Geographic trends: The United States is driving most of the global SELL transactions (389 in total), followed by the UK (130) and China (112). Because of this, the Americas region has the highest volume of total units bought, reaching 37,026 units.